



**Laboratorium
Multimedia dan Internet of Things
Departemen Teknik Komputer
*Institut Teknologi Sepuluh Nopember***

Laporan Akhir Praktikum Jaringan Komputer

Firewall dan NAT

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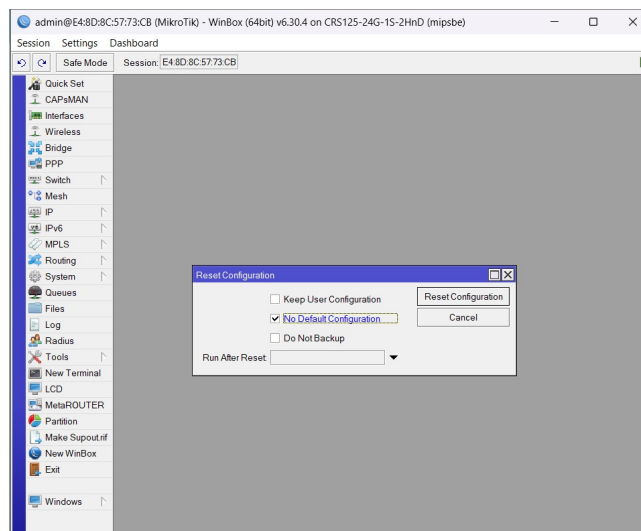
2026

1 Langkah-Langkah Percobaan

1.1 Praktikum 1: Firewall NAT Dasar

1.1.1 Reset Konfigurasi Router

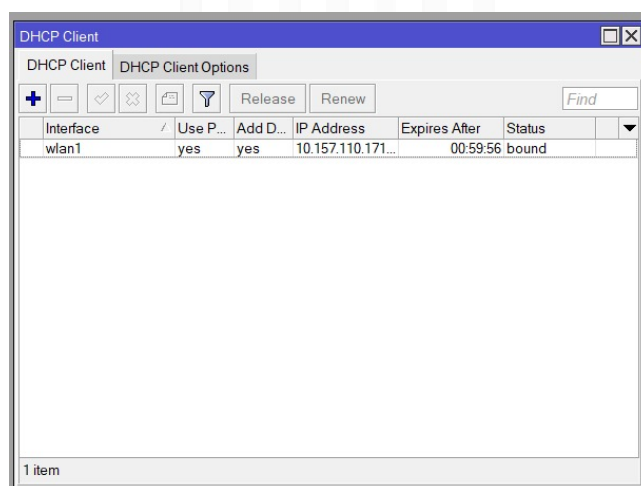
Pastikan router telah di-reset ke kondisi awal tanpa konfigurasi menggunakan aplikasi Winbox pada menu System lalu Reset Configuration dengan mencentang kotak No Default Configuration agar tidak terjadi konflik pada konfigurasi berikutnya.



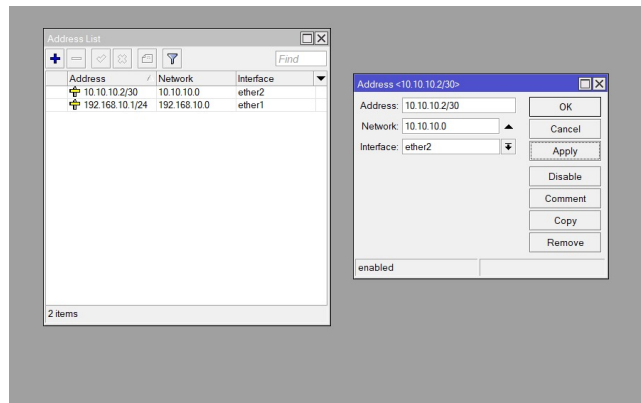
Gambar 1: Reset Router

1.1.2 Konfigurasi DHCP Client pada Router A (ether1)

Sambungkan port ether1 pada Router A ke jaringan Lab atau Internet, lalu tambahkan entri baru pada DHCP Client melalui menu IP untuk ether1 agar router mendapat IP otomatis dan pastikan status koneksinya berubah menjadi bound.



Gambar 2: Konfigurasi DHCP



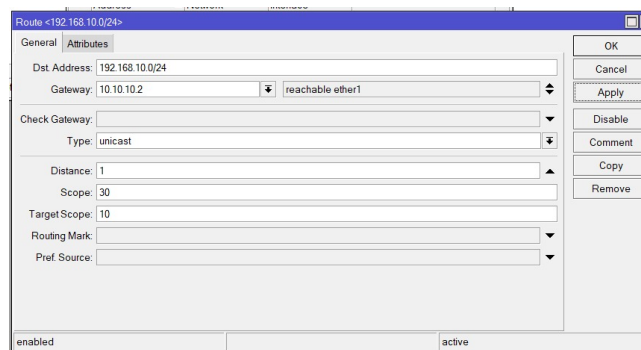
Gambar 3: Konfigurasi IP Address

1.1.3 Konfigurasi IP Address di Router A (ether2)

Tambahkan IP Address 10.10.10.1/30 pada interface ether2 Router A melalui menu IP lalu Addresses sebagai jalur penghubung ke Router B.

1.1.4 Konfigurasi Routing Statis di Router A

Atur routing statis pada Router A melalui menu IP lalu Routes dengan mengisi Dst. Address 192.168.10.0/24 dan Gateway 10.10.10.2 agar Router A mengetahui rute menuju jaringan lokal PC Client yang berada di belakang Router B.



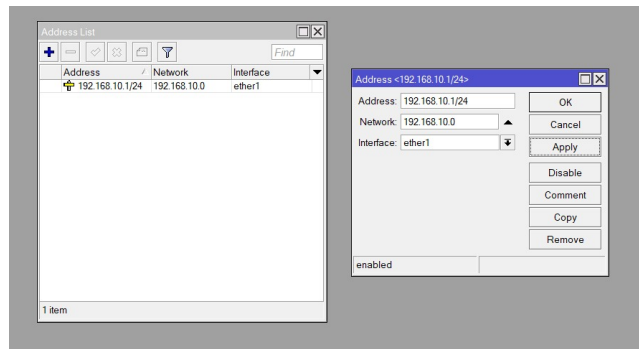
Gambar 4: Konfigurasi Routing Statis

1.1.5 Konfigurasi IP Address di Router B (ether2)

Pada Router B, tambahkan IP Address 192.168.10.1/24 di interface ether2 yang mengarah ke PC Client melalui menu IP lalu Addresses, serta abaikan IP pada ether3.

1.1.6 Konfigurasi IP Address di Router B (ether1)

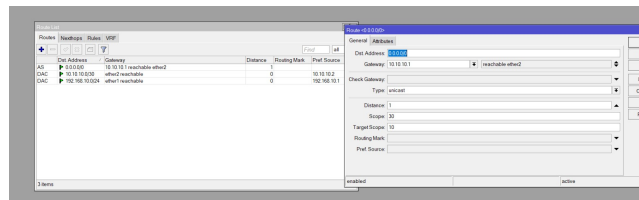
Tambahkan juga IP Address 10.10.10.2/30 pada interface ether1 Router B yang mengarah ke Router A melalui menu IP lalu Addresses.



Gambar 5: Konfigurasi IP Address

1.1.7 Konfigurasi Default Route di Router B

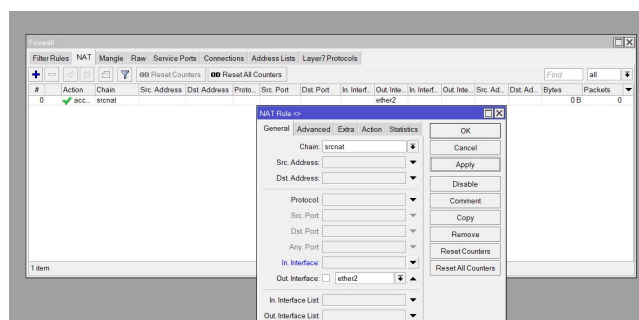
Atur default route pada Router B melalui menu IP lalu Routes dengan membiarkan Dst. Address tetap 0.0.0.0/0 dan mengisi Gateway 10.10.10.1 agar semua paket dari jaringan lokal Router B dikirim ke Router A menuju internet.



Gambar 6: Konfigurasi Default

1.1.8 Konfigurasi Firewall NAT di Router B

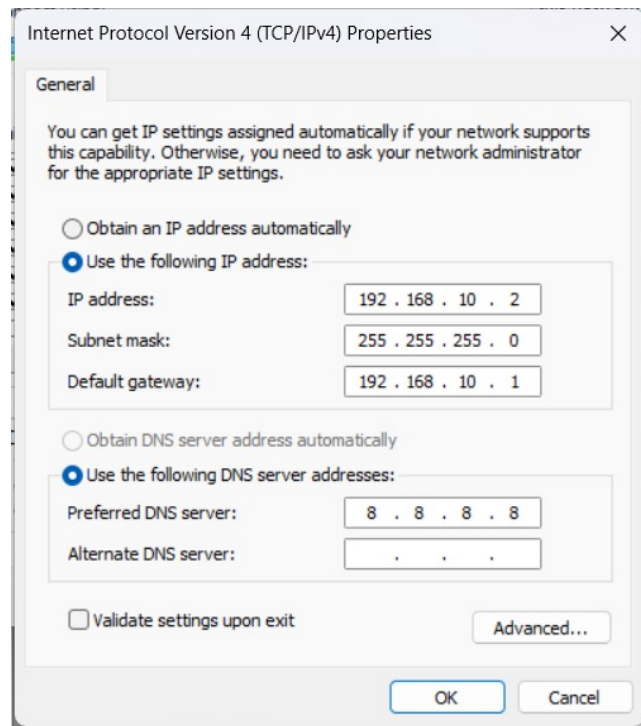
Buat aturan NAT (Network Address Translation) pada Router B melalui menu IP lalu Firewall di tab NAT dengan mengubah Chain menjadi srcnat, Out. Interface menjadi ether1, dan Action menjadi masquerade agar jaringan lokal bisa ikut internetan menggunakan IP publik router.



Gambar 7: Konfigurasi Firewall

1.1.9 Konfigurasi IP Statis di PC Client

Atur IP statis pada PC Client menjadi 192.168.10.2 dengan subnet mask 255.255.255.0 dan default gateway 192.168.10.1 serta DNS 8.8.8.8 melalui pengaturan Network di Control Panel agar bisa terhubung ke Router B.



Gambar 8: Konfigurasi IP

1.1.10 Pengujian Jaringan dari PC Client

Lakukan pengujian jaringan dari PC Client dengan membuka CMD dan melakukan ping ke Gateway Router B, Gateway Router A, serta IP internet atau Lab untuk memastikan routing dan NAT sudah berjalan sukses yang ditandai dengan hasil Reply.

```
[admin@MikroTik] > ping 10.10.10.2
  SEQ HOST                      SIZE TTL TIME  STATUS
  0 10.10.10.2                    56 127 timeout
  1 10.10.10.2                    56 127 timeout
  2 10.10.10.2                    56 127 timeout
  3 10.10.10.2                    56 127 timeout
  4 10.10.10.2                    56 127 timeout
  5 10.10.10.2                    56 127 timeout
  6 10.10.10.2                    56 127 timeout
sent=7 received=0 packet-loss=100%

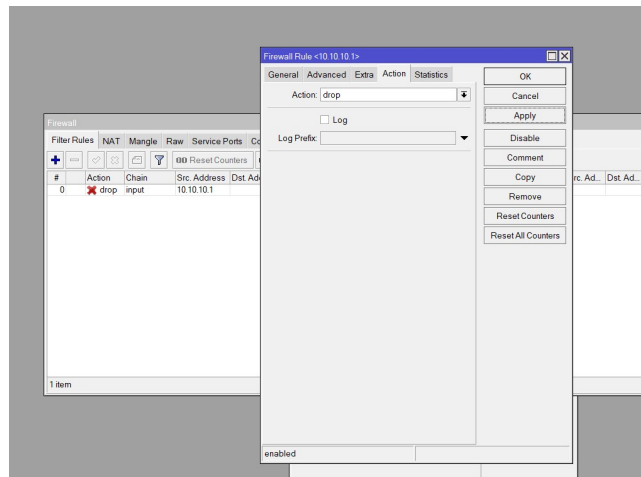
[admin@MikroTik] > ping 192.168.10.2
  SEQ HOST                      SIZE TTL TIME  STATUS
  0 192.168.10.2                  56 127 1ms
  1 192.168.10.2                  56 127 1ms
  2 192.168.10.2                  56 127 1ms
  3 192.168.10.2                  56 127 1ms
  4 192.168.10.2                  56 127 1ms
  5 192.168.10.2                  56 127 1ms
  6 192.168.10.2                  56 127 1ms
  7 192.168.10.2                  56 127 1ms
  8 192.168.10.2                  56 127 1ms
  9 192.168.10.2                  56 127 1ms
 10 192.168.10.2                  56 127 1ms
 11 192.168.10.2                  56 127 1ms
 12 192.168.10.2                  56 127 1ms
 13 192.168.10.2                  56 127 1ms
 14 192.168.10.2                  56 127 1ms
sent=15 received=15 packet-loss=0% min-rtt=1ms avg-rtt=1ms max-rtt=1ms
```

Gambar 9: Pengujian Jaringan

1.2 Praktikum 2: Firewall Filter (Input vs Forward)

1.2.1 Konfigurasi Firewall Filter (Chain Input)

Buka Winbox Router B dan masuk ke menu IP lalu Firewall pada tab Filter Rules untuk membuat aturan baru dengan Chain input, Src. Address 10.10.10.1, Protocol icmp, dan Action drop guna memblokir trafik ping dari Router A yang ditujukan ke Router B.



Gambar 10: Konfigurasi Firewall Filter

1.2.2 Pengujian Chain Input dari Router A

Lakukan pengujian Chain Input dari Router A dengan membuka New Terminal dan melakukan ping ke IP Router B yang hasilnya harus Timeout karena diblokir oleh chain input, serta melakukan ping ke IP PC Client yang hasilnya harus Reply karena paket hanya menumpang lewat Router B.

```
[admin@MikroTik] > ping 10.10.10.2
SEQ HOST                                SIZE TTL TIME STATUS
0 10.10.10.2                            56 127 1ms timeout
1 10.10.10.2                            56 127 1ms timeout
2 10.10.10.2                            56 127 1ms timeout
3 10.10.10.2                            56 127 1ms timeout
4 10.10.10.2                            56 127 1ms timeout
5 10.10.10.2                            56 127 1ms timeout
6 10.10.10.2                            56 127 1ms timeout
sent=7 received=0 packet-loss=100%

[admin@MikroTik] > ping 192.168.10.2
SEQ HOST                                SIZE TTL TIME STATUS
0 192.168.10.2                          56 127 1ms
1 192.168.10.2                          56 127 1ms
2 192.168.10.2                          56 127 1ms
3 192.168.10.2                          56 127 1ms
4 192.168.10.2                          56 127 1ms
```

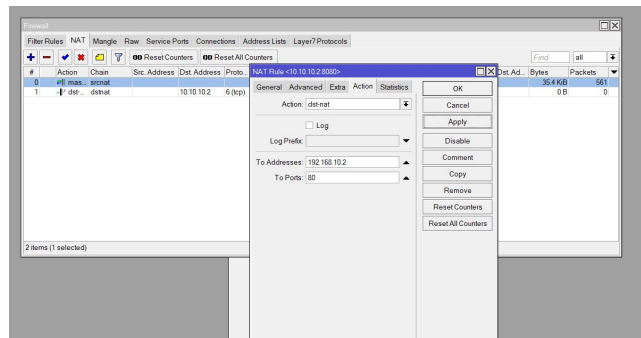
Gambar 11: Pengujian Chain Input

1.2.3 Konfigurasi Firewall Filter (Chain Forward)

Matikan aturan input sebelumnya di Winbox Router B, lalu buat aturan baru dengan Chain forward, Src. Address 10.10.10.1, Protocol icmp, dan Action drop untuk membiarkan Router A melakukan ping ke Router B namun memblokirnya jika Router A melakukan ping ke PC Client.

1.2.4 Pengujian Chain Forward dari Router A

Lakukan pengujian Chain Forward dari Router A dengan melakukan ping ke IP Router B yang hasilnya harus Reply karena aturan input dimatikan, serta melakukan ping ke IP PC Client yang hasilnya harus Timeout karena paket menuju PC Client diblokir oleh chain forward.



Gambar 13: Konfigurasi port Forwarding

```
[admin@MikroTik] > ping 10.10.10.2
SEQ HOST                                SIZE TTL TIME STATUS
0 10.10.10.2                            56 127 1ms timeout
1 10.10.10.2                            56 127 1ms timeout
2 10.10.10.2                            56 127 1ms timeout
3 10.10.10.2                            56 127 1ms timeout
4 10.10.10.2                            56 127 1ms timeout
5 10.10.10.2                            56 127 1ms timeout
6 10.10.10.2                            56 127 1ms timeout
sent=7 received=0 packet-loss=100%

[admin@MikroTik] > ping 192.168.10.2
SEQ HOST                                SIZE TTL TIME STATUS
0 192.168.10.2                          56 127 1ms
1 192.168.10.2                          56 127 1ms
2 192.168.10.2                          56 127 1ms
3 192.168.10.2                          56 127 1ms
4 192.168.10.2                          56 127 1ms
```

Gambar 12: Pengujian Chain Forward

1.3 Praktikum 3: Firewall NAT Forwarding (Port Forwarding)

1.3.1 Konfigurasi Port Forwarding (DstNAT) di Router B

Konfigurasi Port Forwarding (DstNAT) di Router B melalui menu IP lalu Firewall pada tab NAT dengan menambahkan aturan baru berparameter Chain dstnat, Dst. Address 10.10.10.2, Protocol 6 (tcp), Dst. Port 8080, dan Action dst-nat yang diarahkan ke To Addresses 192.168.10.2 pada To Ports 80.

1.3.2 Menjalankan Web Server di PC Client

Jalankan web server sementara di PC Client dengan membuka Command Prompt lalu mengetikkan perintah `python -m http.server 80`, dan biarkan jendela CMD tetap terbuka agar server tetap menyala.

1.3.3 Pengujian Port Forwarding dari Router A

Posisikan Router A sebagai pihak dari luar yang ingin mengakses Web Server di PC Client dengan membuka New Terminal dan mengetikkan perintah `fetch` menuju URL `http://10.10.10.2:8080`, di mana jika berhasil statusnya akan menunjukkan proses connecting dan mengunduh file tanpa error.

1.3.4 Mengamati Connection Tracking

Amati Connection Tracking di Winbox Router B pada tab Connections di dalam menu IP lalu Firewall untuk melihat catatan koneksi dari IP 10.10.10.1 ke 10.10.10.2:8080 yang berstatus established atau syn received, lalu perhatikan respon dari web server PC Client di terminal Router A dan dokumentasikan hasilnya.

```

Terminal<1>
2 192.168.10.2 timeout
3 192.168.10.2 timeout
4 192.168.10.2 timeout
5 192.168.10.2 timeout
6 192.168.10.2 timeout
7 192.168.10.2 timeout
8 192.168.10.2 timeout
sent=9 received=0 packet-loss=100%

[admin@MikroTik] > /tool fetch url="http://10.10.10.2:8080" keep-result=no
failure: invalid URL
[admin@MikroTik] > /tool fetch url="http://10.10.10.2:8080" keep-result=no
[admin@MikroTik] > /tool fetch url="http://10.10.10.2:8080" keep-result=no
failure: invalid URL
[admin@MikroTik] > /tool fetch url="http://10.10.10.2:8080" keep-result=no
failure: invalid URL
[admin@MikroTik] > /tool fetch url="http://10.10.10.2:8080" keep-result=no
failure: invalid URL
[admin@MikroTik] > /tool fetch url="http://10.10.10.2:8080/" keep-result=no
status: failed

failure: closing connection: <connection failed> 10.10.10.2:8080 (4)
[admin@MikroTik] > /tool fetch url="http://10.10.10.2:8080/" keep-result=no\
\... /tool fetch url="http://10.10.10.2:8080/" keep-result=no\
[admin@MikroTik] > /tool fetch url="http://10.10.10.2:8080" keep-result=no
failure: invalid URL
[admin@MikroTik] > /tool fetch url="http://10.10.10.2:8080/" keep-result=no
status: failed

failure: closing connection: <connection failed> 10.10.10.2:8080 (4)
[admin@MikroTik] > /tool fetch url="http://10.10.10.2:8080/" keep-result=no
status: failed

failure: closing connection: <connection failed> 10.10.10.2:8080 (4)
[admin@MikroTik] >

```

Gambar 14: Pengujian Port Forwarding

2 Analisis Hasil Percobaan

Berdasarkan praktikum yang telah dilakukan, dapat dianalisis beberapa prinsip kerja jaringan dan keamanan:

2.1 Fungsi NAT (Masquerade)

Pada Praktikum 1, PC Client dengan IP privat (192.168.10.2) berhasil mengakses jaringan luar/internet karena fungsi Source NAT (masquerade) pada Router B. Router B bertugas menerjemahkan alamat IP lokal menjadi alamat publik/luar yang terdaftar di antarmuka routernya, sehingga balasan paket data dapat dikembalikan ke rute yang tepat.

2.2 Chain Input vs Forward

Praktikum 2 secara jelas membuktikan perbedaan pengelolaan lalu lintas data. Chain Input menangani paket yang tujuan akhirnya adalah router itu sendiri (sehingga ping ke Router B gagal saat di-drop). Sebaliknya, Chain Forward mengelola paket yang hanya melewati router (numpang lewat). Saat aturan forward di-drop, Router A bisa nge-ping router B, tetapi tidak bisa nge-ping PC Client.

2.3 Destination NAT (Port Forwarding)

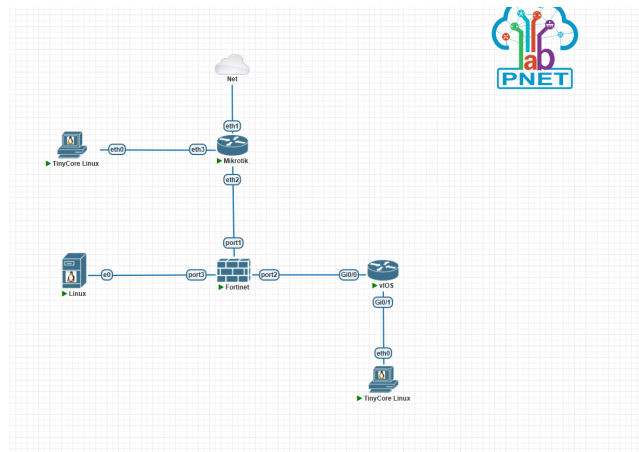
Praktikum 3 menunjukkan bahwa perangkat luar tidak bisa menjangkau IP privat secara langsung. Dengan DstNAT, permintaan ke IP luar router pada port tertentu (8080) secara otomatis diarahkan ke IP dan port lokal (80) tempat server berjalan. Hal ini sangat efisien untuk melindungi server internal tanpa harus mengekspos seluruh jaringan ke publik.

2.4 Connection Tracking

Melalui pengamatan stateful inspection, router terbukti mencatat parameter Source/Destination Address dan Port. Hal ini memungkinkan router membedakan mana koneksi sah yang sedang berjalan (established) dan mana paket yang tidak dikenal.

3 Hasil Tugas Modul

3.1 Topologi Jaringan



Gambar 15: Topologi Jaringan

3.2 Tabel IP Address

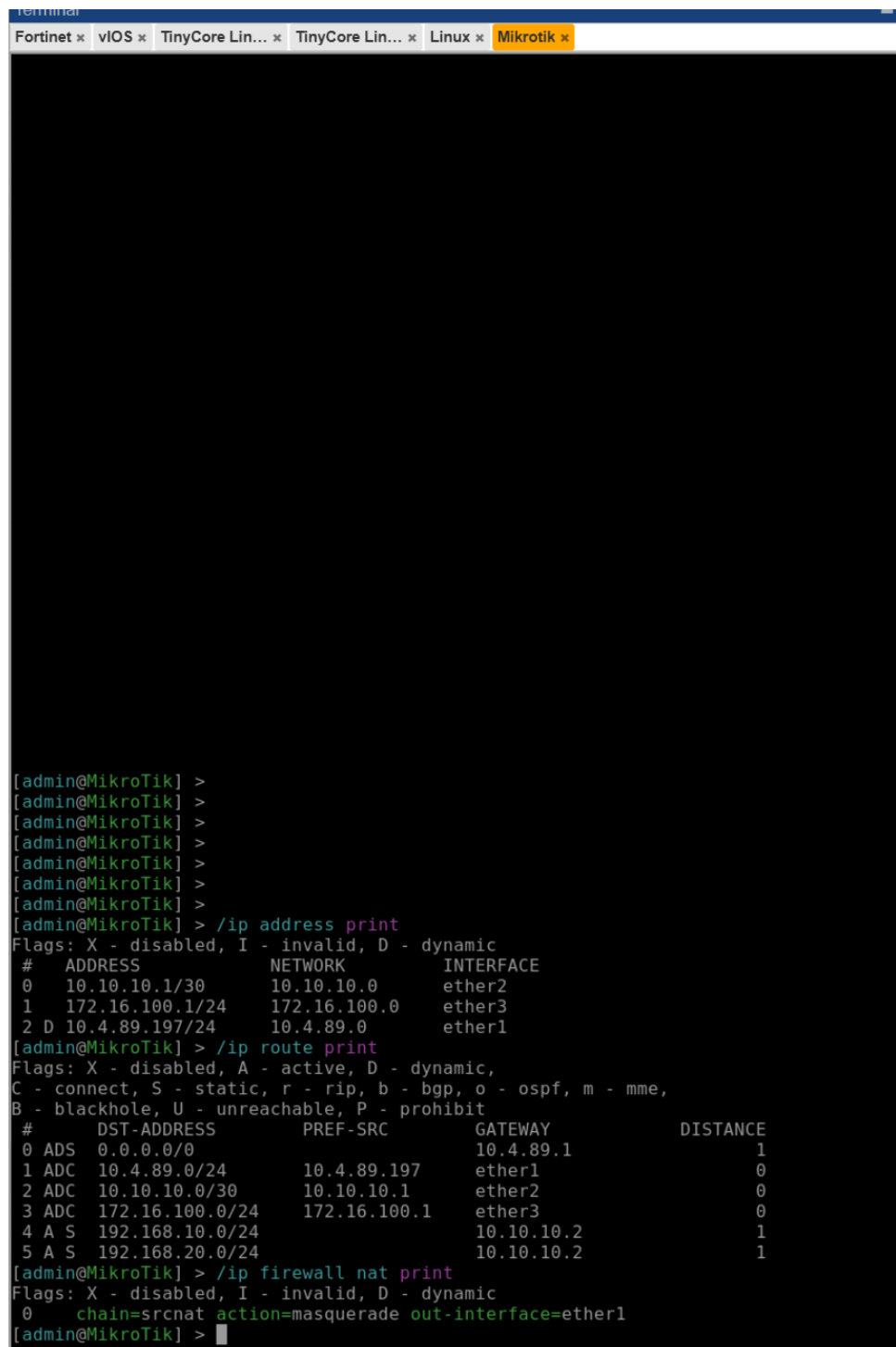
Pengalamatan IP pada setiap perangkat dikonfigurasi berdasarkan rancangan segmentasi jaringan berikut:

Tabel 1: Tabel Konfigurasi IP Address

Perangkat	Interface	IP Address	Gateway
MikroTik ISP	ether1	DHCP Client	DHCP Lab
MikroTik ISP	ether2	10.10.10.1/30	-
MikroTik ISP	ether3	172.16.100.1/24	-
FortiGate	port1	10.10.10.2/30	10.10.10.1
FortiGate	port2	10.20.20.1/30	-
FortiGate	port3	192.168.20.1/24	-
Cisco Router	G0/0	10.20.20.2/30	-
Cisco Router	G0/1	192.168.10.1/24	-
Client LAN	eth0	192.168.10.10/24	192.168.10.1
Client WAN	eth0	172.16.100.10/24	172.16.100.1
Ubuntu DMZ	eth0	192.168.20.10/24	192.168.20.1

3.3 Konfigurasi Tiap Perangkat

3.3.1 Konfigurasi MikroTik ISP



```
terminal
Fortinet x vIOS x TinyCore Lin... x TinyCore Lin... x Linux x Mikrotik x

[admin@MikroTik] >
[admin@MikroTik] >
[admin@MikroTik] >
[admin@MikroTik] >
[admin@MikroTik] >
[admin@MikroTik] >
[admin@MikroTik] >
[admin@MikroTik] > /ip address print
Flags: X - disabled, I - invalid, D - dynamic
# ADDRESS NETWORK INTERFACE
0 10.10.10.1/30 10.10.10.0 ether2
1 172.16.100.1/24 172.16.100.0 ether3
2 D 10.4.89.197/24 10.4.89.0 ether1
[admin@MikroTik] > /ip route print
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS PREF-SRC GATEWAY DISTANCE
0 ADS 0.0.0.0/0 10.4.89.1 1
1 ADC 10.4.89.0/24 10.4.89.197 ether1 0
2 ADC 10.10.10.0/30 10.10.10.1 ether2 0
3 ADC 172.16.100.0/24 172.16.100.1 ether3 0
4 A S 192.168.10.0/24 10.10.10.2 1
5 A S 192.168.20.0/24 10.10.10.2 1
[admin@MikroTik] > /ip firewall nat print
Flags: X - disabled, I - invalid, D - dynamic
0 chain=srcnat action=masquerade out-interface=ether1
[admin@MikroTik] >
```

Gambar 16: Konfigurasi Mikrotik

3.3.2 Konfigurasi FortiGate

```
FortiGate-VM64-KVM # show system interface
config system interface
  edit "port1"
    set vdom "root"
    set ip 10.10.10.2 255.255.255.252
    set allowaccess ping
    set type physical
    set snmp-index 1
  next
  edit "port2"
    set vdom "root"
    set ip 10.20.20.1 255.255.255.252
    set allowaccess ping
    set type physical
    set snmp-index 2
  next
  edit "port3"
    set vdom "root"
    set ip 192.168.20.1 255.255.255.0
    set allowaccess ping
    set type physical
    set snmp-index 3
  next
  edit "port4"
    set vdom "root"
    set type physical
  next
FortiGate-VM64-KVM # show router static
config router static
  edit 1
    set gateway 10.10.10.1
    set device "port1"
  next
  edit 2
    set dst 192.168.10.0 255.255.255.0
    set gateway 10.20.20.2
    set device "port2"
  next
end
```

Gambar 17: Show Interface dan show router static

```
terminal
Fortinet x vIOS x TinyCore Lin... x TinyCore Lin... x Linux x
end

FortiGate-VM64-KVM # show firewall address
config firewall address
  edit "none"
    set uuid 93f784ea-61ce-51f1-6561-e8b5b959025b
    set subnet 0.0.0.0 255.255.255.255
  next
  edit "login.microsoftonline.com"
    set uuid 93f78c24-61ce-51f1-0923-be93489d89ed
    set type fqdn
    set fqdn "login.microsoftonline.com"
  next
  edit "login.microsoft.com"
    set uuid 93f7916a-61ce-51f1-41e6-f61a8feedbbf
    set type fqdn
    set fqdn "login.microsoft.com"
  next
  edit "login.windows.net"
    set uuid 93f79426-61ce-51f1-3d67-b0773f9d94ce
    set type fqdn
    set fqdn "login.windows.net"
  next
  edit "gmail.com"
    set uuid 93f79700-61ce-51f1-0acf-4d2b5b10a25f
    set type fqdn
    set fqdn "gmail.com"
  next
  edit "wildcard.google.com"
    set uuid 93f79958-61ce-51f1-5437-33f1620a213a
    set type fqdn
    set fqdn "*.google.com"
  next
  edit "wildcard.dropbox.com"
    set uuid 93f79bf6-61ce-51f1-822b-f6365026f420
    set type fqdn
    set fqdn "*.dropbox.com"
  next
  edit "all"
    set uuid 9495ae5e-61ce-51f1-b14b-a63be0d74e4e
  next
  edit "FIREWALL_AUTH_PORTAL_ADDRESS"
    set uuid 9495b35e-61ce-51f1-1296-f39bf4b6207a
  next
  edit "FABRIC_DEVICE"
    set uuid 9495b476-61ce-51f1-137b-6cbbef70bae4
    set comment "IPv4 addresses of Fabric Devices."
  next
  edit "SSLVPN_TUNNEL_ADDR1"
    set uuid 94962e60-61ce-51f1-0991-32b6afef713c
    set type iprange
    set start-ip 10.212.134.200
    set end-ip 10.212.134.210
  next
  edit "LAN_NETWORK"
    set uuid cc540e8a-6214-51f1-f546-ae9287ceb036
    set subnet 192.168.10.0 255.255.255.0
  next
  edit "DMZ_Server"
    set uuid 02640200-6215-51f1-2804-61f607698fa7
    set subnet 192.168.20.10 255.255.255.255
```

Gambar 18: Cek Firewall Address

```
terminal
Fortinet x vIOS x TinyCore Lin... x TinyCore Lin... x Linux x
Unknown action 0

FortiGate-VM64-KVM # show firewall policy
config firewall policy
  edit 1
    set name "LAN_to_WAN"
    set uuid 85b8f088-6216-51f1-6e51-50c02b3bf629
    set srcintf "port2"
    set dstintf "port1"
    set action accept
    set srcaddr "LAN_NETWORK"
    set dstaddr "all"
    set schedule "always"
    set service "ALL"
    set nat enable
  next
  edit 2
    set name "LAN_to_DMZ"
    set uuid 09666e56-6217-51f1-8c7a-fbeba7927fc4
    set srcintf "port2"
    set dstintf "port3"
    set action accept
    set srcaddr "LAN_NETWORK"
    set dstaddr "DMZ_Server"
    set schedule "always"
    set service "ALL"
  next
  edit 3
    set name "WAN_to_DMZ_HTTP"
    set uuid b388b5a4-6219-51f1-95f8-1a41205a5810
    set srcintf "port1"
    set dstintf "port3"
    set action accept
    set srcaddr "WAN_Client_Network"
    set dstaddr "VIP_HTTP_DMZ"
    set schedule "always"
    set service "ALL"
  next
  edit 4
    set name "DMZ_to_Internet"
    set uuid 6561ed68-621f-51f1-0481-4ef584762539
    set srcintf "port3"
    set dstintf "port1"
    set action accept
    set srcaddr "DMZ_Server"
    set dstaddr "all"
    set schedule "always"
    set service "ALL"
    set nat enable
  next
end

FortiGate-VM64-KVM # show firewall vip
config firewall vip
  edit "VIP_DMZ_HTTP"
    set uuid 44645fb0-6215-51f1-001b-56590dbf9d3c
    set extip 10.10.10.2
    set mappedip "192.168.20.10"
    set extintf "port1"
    set portforward enable
    set extport 80
    set mappedport 80
  next
```

Gambar 19: Show firewall Policy dan Vip

3.3.3 Konfigurasi Cisco vIOS Router

```
terminal
Fortinet x VIOS x TinyCore Lin... x TinyCore Lin... x Linux x
!
multilink bundle-name authenticated
!
!
!
!
redundancy
!
!
interface GigabitEthernet0/0
 ip address 10.20.20.2 255.255.255.252
 duplex auto
 speed auto
 media-type rj45
>
interface GigabitEthernet0/1
 ip address 192.168.10.1 255.255.255.0
 duplex auto
 speed auto
 media-type rj45
!
ip forward-protocol nd
!
no ip http server
no ip http secure-server
ip route 0.0.0.0 0.0.0.0 10.20.20.1
!
ipv6 ioam timestamp
!
control-plane
!
banner exec ^C
*****
* IOSv is strictly limited to use for evaluation, demonstration and IOS *
* education. IOSv is provided as-is and is not supported by Cisco's   *
* Technical Advisory Center. Any use or disclosure, in whole or in part,*
* of the IOSv Software or Documentation to any third party for any    *
* purposes is expressly prohibited except as otherwise authorized by  *
* Cisco in writing.                                                    *
*****^C
banner incoming ^C
*****
* IOSv is strictly limited to use for evaluation, demonstration and IOS *
* education. IOSv is provided as-is and is not supported by Cisco's   *
* Technical Advisory Center. Any use or disclosure, in whole or in part,*
```

Gambar 20: Konfigurasi Cisco vIOS

3.3.4 Konfigurasi Ubuntu Server (DMZ)



```
Terminal
Fortinet x vIOS x TinyCore Lin... x TinyCore Lin... x Linux x Mikrotik x

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

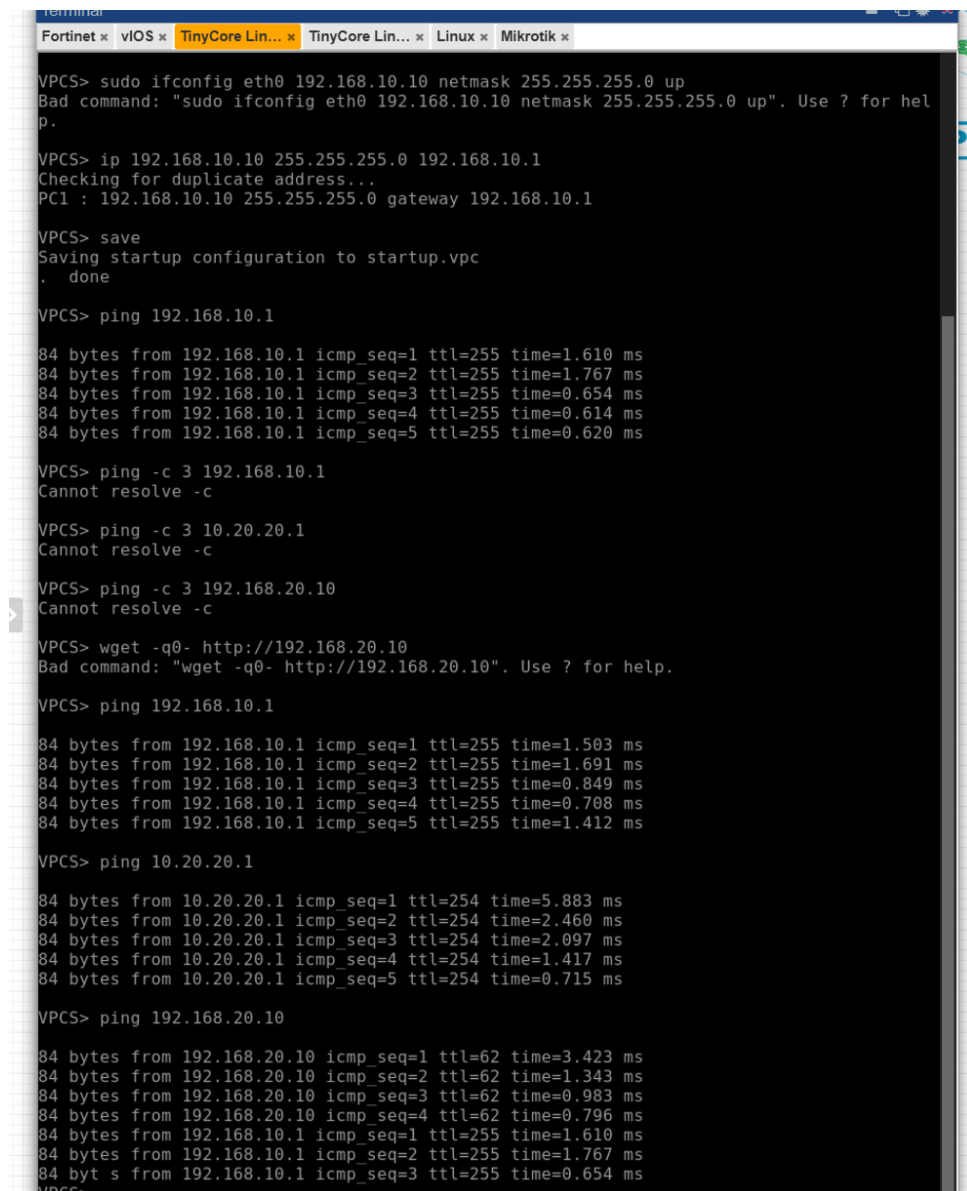
<p><em>Thank you for using nginx.</em></p>
</body>
</html>
root@kum:~# echo "Tunod_4_DMZ_Firewall_27-Peer to Peer" > /var/www/html/index.html
root@kum:~# curl http://192.168.20.10
Tunod_4_DMZ_Firewall_27-Peer to Peer
root@kum:~# ip addr show eth0
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 50:56:1e:00:b5:00 brd ff:ff:ff:ff:ff:ff
    inet 192.168.20.10/24 brd 192.168.20.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::5256:1eff:fe00:b500/64 scope link
        valid_lft forever preferred_lft forever
root@kum:~# cat /etc/netplan/00-installer-config.yaml
network:
  ethernet:
    eth0:
      dhcp4: no
      addresses:
        - 192.168.20.10/24
      gateway4: 192.168.20.1
      nameservers:
        addresses: [8.8.8.8]
      version: 2
root@kum:~# systemctl status nginx
* nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/lib/systemd/system/nginx.service; enabled; vendor preset: enabled)
   Active: active (running) since Sun 2026-06-07 12:28:19 CST; 54min ago
     Docs: man:nginx(8)
    Main PID: 1981 (nginx)
      Tasks: 9 (limit: 4620)
     Memory: 5.7M
    CGroup: /system.slice/nginx.service
            └─1981 nginx: master process /usr/sbin/nginx -g daemon on: master_process on:
               └─1982 nginx: worker process
                  └─1983 nginx: worker process

Jun 07 12:28:19 kum systemd[1]: Starting A high performance web server and a reverse proxy server...
Jun 07 12:28:19 kum systemd[1]: Started A high performance web server and a reverse proxy server.
root@kum:~# curl http://192.168.20.10
Tunod_4_DMZ_Firewall_27-Peer to Peer
root@kum:~#
```

Gambar 21: Cek Linux

3.4 Hasil Pengujian

3.4.1 Pengujian dari PC LAN



```
Fortinet x VIOS x TinyCore Lin... x TinyCore Lin... x Linux x Mikrotik x

VPCS> sudo ifconfig eth0 192.168.10.10 netmask 255.255.255.0 up
Bad command: "sudo ifconfig eth0 192.168.10.10 netmask 255.255.255.0 up". Use ? for help.

VPCS> ip 192.168.10.10 255.255.255.0 192.168.10.1
Checking for duplicate address...
PC1 : 192.168.10.10 255.255.255.0 gateway 192.168.10.1

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> ping 192.168.10.1

84 bytes from 192.168.10.1 icmp_seq=1 ttl=255 time=1.610 ms
84 bytes from 192.168.10.1 icmp_seq=2 ttl=255 time=1.767 ms
84 bytes from 192.168.10.1 icmp_seq=3 ttl=255 time=0.654 ms
84 bytes from 192.168.10.1 icmp_seq=4 ttl=255 time=0.614 ms
84 bytes from 192.168.10.1 icmp_seq=5 ttl=255 time=0.620 ms

VPCS> ping -c 3 192.168.10.1
Cannot resolve -c

VPCS> ping -c 3 10.20.20.1
Cannot resolve -c

VPCS> ping -c 3 192.168.20.10
Cannot resolve -c

VPCS> wget -q0- http://192.168.20.10
Bad command: "wget -q0- http://192.168.20.10". Use ? for help.

VPCS> ping 192.168.10.1

84 bytes from 192.168.10.1 icmp_seq=1 ttl=255 time=1.503 ms
84 bytes from 192.168.10.1 icmp_seq=2 ttl=255 time=1.691 ms
84 bytes from 192.168.10.1 icmp_seq=3 ttl=255 time=0.849 ms
84 bytes from 192.168.10.1 icmp_seq=4 ttl=255 time=0.708 ms
84 bytes from 192.168.10.1 icmp_seq=5 ttl=255 time=1.412 ms

VPCS> ping 10.20.20.1

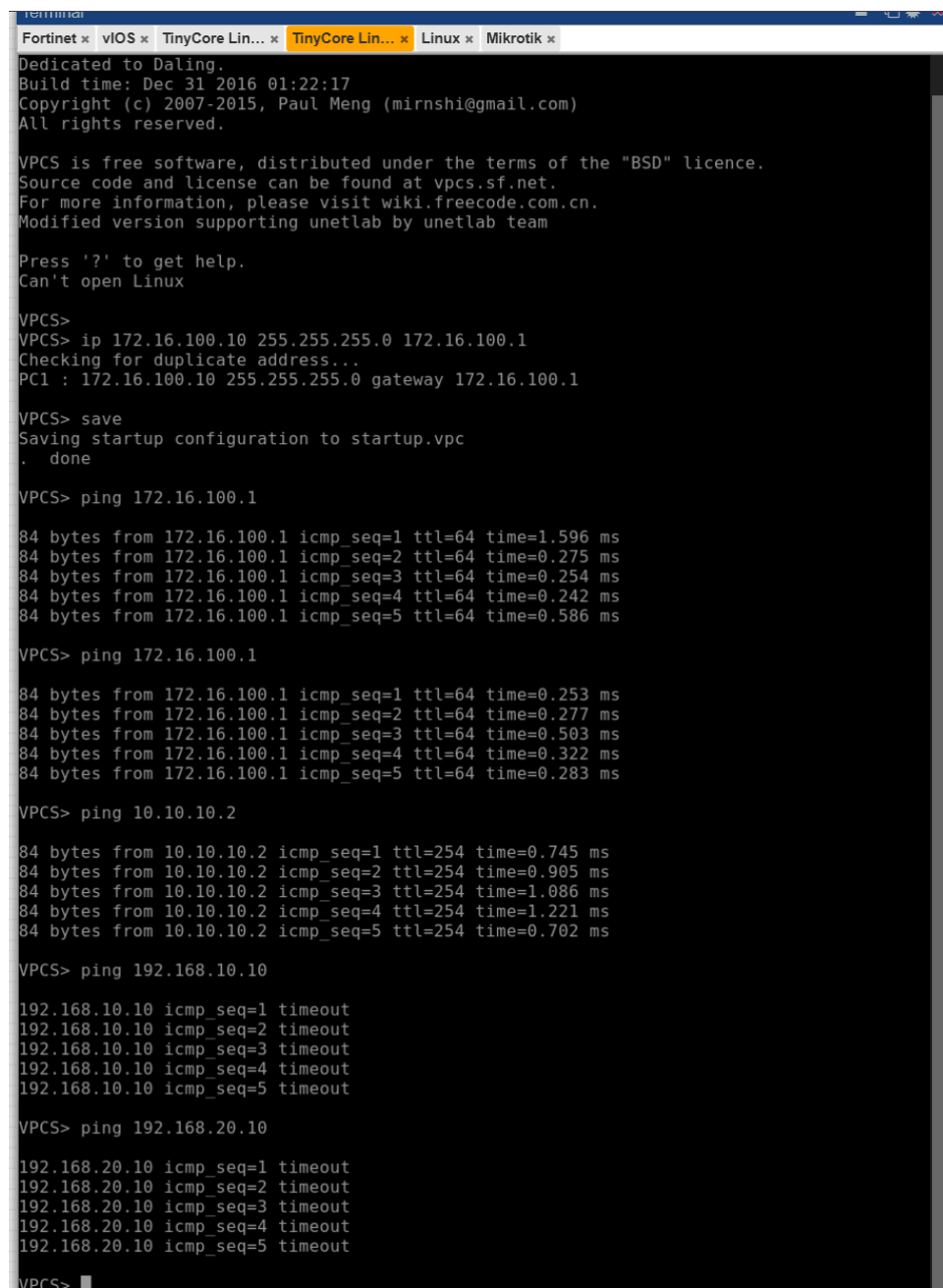
84 bytes from 10.20.20.1 icmp_seq=1 ttl=254 time=5.883 ms
84 bytes from 10.20.20.1 icmp_seq=2 ttl=254 time=2.460 ms
84 bytes from 10.20.20.1 icmp_seq=3 ttl=254 time=2.097 ms
84 bytes from 10.20.20.1 icmp_seq=4 ttl=254 time=1.417 ms
84 bytes from 10.20.20.1 icmp_seq=5 ttl=254 time=0.715 ms

VPCS> ping 192.168.20.10

84 bytes from 192.168.20.10 icmp_seq=1 ttl=62 time=3.423 ms
84 bytes from 192.168.20.10 icmp_seq=2 ttl=62 time=1.343 ms
84 bytes from 192.168.20.10 icmp_seq=3 ttl=62 time=0.983 ms
84 bytes from 192.168.20.10 icmp_seq=4 ttl=62 time=0.796 ms
84 bytes from 192.168.10.1 icmp_seq=1 ttl=255 time=1.610 ms
84 bytes from 192.168.10.1 icmp_seq=2 ttl=255 time=1.767 ms
84 bytes from 192.168.10.1 icmp_seq=3 ttl=255 time=0.654 ms
VPCS>
```

Gambar 22: Pengujian pc LAN

3.4.2 Pengujian dari PC WAN



```
Fortinet x vIOS x TinyCore Lin... x TinyCore Lin... x Linux x Mikrotik x
Dedicated to Daling.
Build time: Dec 31 2016 01:22:17
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.
Modified version supporting unetlab by unetlab team

Press '?' to get help.
Can't open Linux

VPCS>
VPCS> ip 172.16.100.10 255.255.255.0 172.16.100.1
Checking for duplicate address...
PC1 : 172.16.100.10 255.255.255.0 gateway 172.16.100.1

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> ping 172.16.100.1

84 bytes from 172.16.100.1 icmp_seq=1 ttl=64 time=1.596 ms
84 bytes from 172.16.100.1 icmp_seq=2 ttl=64 time=0.275 ms
84 bytes from 172.16.100.1 icmp_seq=3 ttl=64 time=0.254 ms
84 bytes from 172.16.100.1 icmp_seq=4 ttl=64 time=0.242 ms
84 bytes from 172.16.100.1 icmp_seq=5 ttl=64 time=0.586 ms

VPCS> ping 172.16.100.1

84 bytes from 172.16.100.1 icmp_seq=1 ttl=64 time=0.253 ms
84 bytes from 172.16.100.1 icmp_seq=2 ttl=64 time=0.277 ms
84 bytes from 172.16.100.1 icmp_seq=3 ttl=64 time=0.503 ms
84 bytes from 172.16.100.1 icmp_seq=4 ttl=64 time=0.322 ms
84 bytes from 172.16.100.1 icmp_seq=5 ttl=64 time=0.283 ms

VPCS> ping 10.10.10.2

84 bytes from 10.10.10.2 icmp_seq=1 ttl=254 time=0.745 ms
84 bytes from 10.10.10.2 icmp_seq=2 ttl=254 time=0.905 ms
84 bytes from 10.10.10.2 icmp_seq=3 ttl=254 time=1.086 ms
84 bytes from 10.10.10.2 icmp_seq=4 ttl=254 time=1.221 ms
84 bytes from 10.10.10.2 icmp_seq=5 ttl=254 time=0.702 ms

VPCS> ping 192.168.10.10

192.168.10.10 icmp_seq=1 timeout
192.168.10.10 icmp_seq=2 timeout
192.168.10.10 icmp_seq=3 timeout
192.168.10.10 icmp_seq=4 timeout
192.168.10.10 icmp_seq=5 timeout

VPCS> ping 192.168.20.10

192.168.20.10 icmp_seq=1 timeout
192.168.20.10 icmp_seq=2 timeout
192.168.20.10 icmp_seq=3 timeout
192.168.20.10 icmp_seq=4 timeout
192.168.20.10 icmp_seq=5 timeout

VPCS>
```

Gambar 23: Pengujian pc WAN

3.5 Analisis dan Kesimpulan

Berdasarkan pengerjaan tugas modul, dapat disimpulkan bahwa arsitektur jaringan yang menerapkan konsep *Demilitarized Zone* (DMZ) menggunakan *firewall* (FortiGate) memberikan tingkat keamanan dan segmentasi yang sangat baik.

Pemisahan zona menjadi WAN, LAN, dan DMZ memastikan bahwa meskipun server publik (di DMZ) dapat diakses oleh dunia luar (melalui *Destination NAT / VIP*), jaringan LAN internal tetap terisolasi dan tidak bisa diakses secara langsung oleh pihak luar (seperti yang dibuktikan oleh status **timeout** saat PC WAN mencoba melakukan ping ke jaringan LAN). Penggunaan *Virtual IP* juga terbukti efektif dalam menyembunyikan alamat IP asli dari server DMZ, sehingga mencegah potensi serangan jaringan yang menargetkan IP lokal secara langsung.

4 Kesimpulan

Praktikum Modul 4 berhasil mengimplementasikan arsitektur jaringan yang aman melalui segmentasi WAN, LAN, dan DMZ menggunakan firewall FortiGate. Sesuai dengan teori keamanan jaringan, hasil pengujian membuktikan bahwa jaringan internal LAN dan IP lokal DMZ berhasil diisolasi secara total dari ancaman luar. Meskipun demikian, layanan web server pada zona DMZ tetap dapat diakses oleh publik secara aman melalui mekanisme Virtual IP (Port Forwarding). Dari praktikum ini, praktikan berhasil memahami penerapan Firewall Policy untuk mengatur hak akses antar-zona, pentingnya menyembunyikan IP privat, serta memperoleh keterampilan teknis dalam mengintegrasikan perangkat lintas vendor seperti MikroTik, FortiGate, dan Cisco.

5 Lampiran



Gambar 24: Lampiran



Gambar 25: Lampiran